



COMPUTER SCIENCE AND SOCIETY

AGENDA

Computer Science, Society, and the Information Society

1

The Role of Data in the Information Society

2

Economic Impact of the Information Society

3

Social Impacts of the Information Society

4

Infrastructure Vulnerability, Computer Science and the Military

5

Responsibility in Computer Science

6

UNIT2

THE ROLE OF DATA IN THE INFORMATION SOCIETY



On completion of this unit, you will be able to...

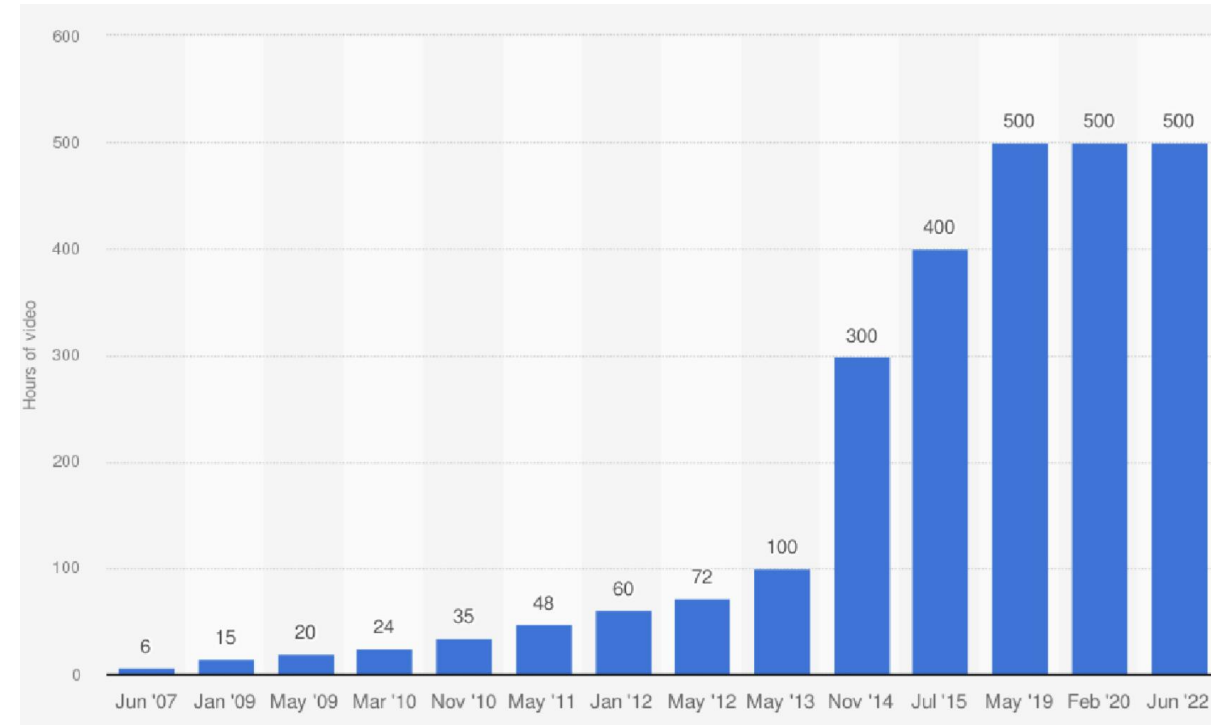
- name **hazards** of the information society.
- understand crucial aspects of **data generation** and **data protection**.
- describe the functioning of industry-shaping systems like **ERP systems** or **cyber-physical systems** (IoT).
- discuss data as a **commodity**.



1. Explain why **data** has been called “**the new oil**”
2. Explain what is meant by “**Industry 4.0**”
3. Name a **law** regarding **data privacy**.

THE AMOUNT OF DATA IS GROWING EXPONENTIALLY

- 2025: 181 zettabytes (1 ZB = $1 \cdot 10^{21}$ bytes)
- e.g. **YouTube**: now 500 hours of video content uploaded **per minute**



YES

- an immensely, untapped **valuable asset**
- those who see Data's fundamental value and learn to extract and use it there will be **huge rewards**"

(Toonders, 2014)

NO

- potentially **infinitely available**
- can be **copied** at almost no marginal cost
- “much **more valuable** than oil or gold, namely infinitely valuable” (Bendig, 2020).

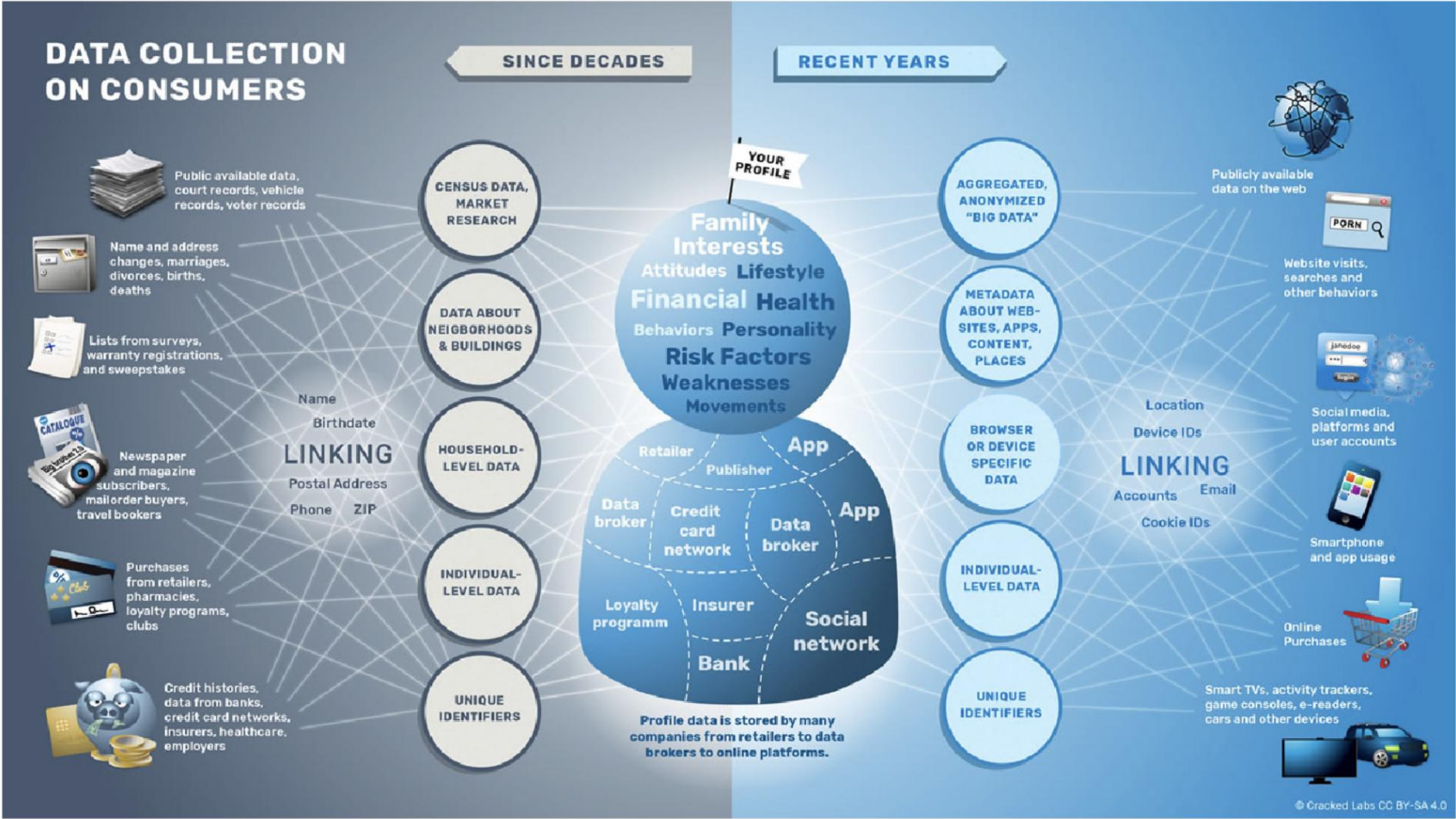
Enterprise Resource Planning

- ERP systems cover finance and accounting, human resources, materials management, logistics, production, and sales data
- Data is collected and made accessible for **analysis**

Sensors

- sensors in **industrial production** continuously generate data
- E.g. **RFID** tags (Radio Frequency Identification) to track individual parts
- Digitalized production processes: cyber-physical systems
- „**Industry 4.0**“ in Germany (success story!)

DATA ON INDIVIDUALS



Source: Cracked Labs (2017). CC BY-SA 4.0.

VALUE OF PERSONAL DATA

- Personal data powers targeted advertising: Meta reported USD 164.5 billion revenue in 2024, mainly from advertising.
- Google/Alphabet generated about USD 350 billion total revenue in 2024; advertising remained the largest revenue source.



<https://www.visualcapitalist.com/alphabets-revenue-breakdown-in-2024/>

<https://investor.atmeta.com/investor-news/press-release-details/2025/Meta-Reports-Fourth-Quarter-and-Full-Year-2024-Results/default.aspx?os=io...>

- identifying **patterns** in vast data sets
- users analysed based on their **previous online behaviour**
- patterns can be used to **tailor advertising**
- Ability to precisely address a **target group** or even an individual

- Protecting personal data from **misuse**
- Strong protection in the EU EU: **General Data Protection Regulation**, GDPR, 2016
- Severe **penalties** for violations
- Also affected standards in the US

- **Cyber spying:** intruding into private/ enterprise hard drives or servers to obtain data
- Tools: e.g.
 - proxy servers
 - code cracking methods
 - **spyware**



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SESSION 3

TRANSFER TASK

Case Study: Google Ads

How (and why) can Google Ads help **small businesses** extend their business?

1. Spend some time working through the **help materials** for new customers of Google Ads, e.g.:
 - Start page: https://ads.google.com/intl/en_uk/home/education/
 - Help topics, e.g. “start advertising”: <https://support.google.com/google-ads/?hl=en-GB&sjid=12997424877365188434-EU#topic=10286612>

TRANSFER TASK

1. Describe **what data is used** in setting up and running an ad campaign!
How is it used?
2. What are the **advantages and disadvantages** of using Google Ads?

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your
results.

The results will be
discussed in plenary.





1. „Data is the new...”

- a) Gold
- b) Coal
- c) Oil
- d) Software



2. Which area is **NOT** covered by ERP (Enterprise Resource Planning) systems?

- a) Project management
- b) Logistics
- c) Sales
- d) Human resources

LIST OF SOURCES

Bendig, T. (2020, November 17). Viel wertvoller als Öl und Gold [Much more worthwhile than gold]. Fraunhofer InnoVisions. <https://www.fraunhofer-innovisions.de/big-data/l-ebendige-zukunft>

Cracked Labs (2017). CC BY-SA 4.0.

Kleppmann, M. (2017): *Designing data-intensive applications. The big ideas behind reliable, scalable, and maintainable systems*. 1st Edition. Sebastopol, CA: O'Reilly.

Statista. (2023). Hours of Video Uploaded to YouTube Every Minute as of February 2022. <https://www-statista-com.pxz.iubh.de:8443/statistics/259477/hours-of-video-uploaded-to-youtube-every-minute>

Toonders, J. (2014, July 23). Data Is the New Oil of the Digital Economy. Wired. Retrieved from <https://www.wired.com/insights/2014/07/data-new-oil-digital-economy/>

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